THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(b)** 10 N	As per the textbook values
2	**(d)** force with which Earth pulls it	Measured in newtons
3	**(a)** electrostatic force	Static charges
4	**(c)** in all directions	At bottom and on sides
5	**(a)** height of the liquid column	Greater depth gives greater pressure
6	**(a)** positive charges	Lighter ice particles
7	**(c)** vibrate more vigorously	Energy gain
8	**(b)** large interparticle spaces exist	Particles come closer
9	**(a)** two or more metals	Properties different from the metals
10	**(a)** mixed without chemical combination	Components keep their properties
11	**(c)** 30 g	250 − 220
12	**(c)** decreases	Fizz escapes from warm soda

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	They repel each other (1). This is electrostatic force (1).	1+1
14	Force per unit area (1). P = F/A (1).	1+1
15	Gas particles are far apart (1). Water particles are close with little space (1).	1+1
16	Simplest substance that cannot be broken down (1). Oxygen and gold (1).	1+1
17	Solute is the dissolved substance, smaller amount (1). Solvent is the medium, larger amount (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	Several forces may act (1). They balance one another (1). Net force is zero (1).	1+1+1
19	Ice: regular, close (1). Water: loosely arranged, mobile (1). Vapour: far apart and free (1).	1+1+1
20	Iron reacts with oxygen and moisture (1). New substance iron oxide forms (1). It has fixed composition (1).	1+1+1
21	Density = 120 ÷ 40 = 3 g/cm³ (1½). More than 1 g/cm³ so it sinks (1½).	1½+1½
22	Components are not chemically bound (1½). Magnet separates iron from sulfur (1½).	1½+1½
	OR: No (1). An element has one kind of matter (1). A compound has two or more elements combined (1).	1+1+1
23	Water particles rearrange on freezing (1). They occupy more space (1). So ice has lower density (1).	1+1+1
	OR: Substance expands on heating (1½). Volume increases while mass is the same (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	Pressure by air (1). Rubber sucker sticking (2). Balloon expanding or bottle crushing (2).	1+2+2
	OR: High speed means low pressure (1). Balloon activity (2). Roof example (2).	1+2+2
25	Take 100 mL water (1). Add salt in portions with stirring (1). Stop when it just stays undissolved (1). Weigh the salt added (1). That mass is the solubility (1).	1+1+1+1+1
	OR: Place egg in tap water, it sinks (1). Add salt gradually with stirring (1). Water density increases (1). Egg floats when solution is denser (1). Sinking again on dilution (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** spring balance	Stretches a spring
2	**(b)** static charges	Build up on rubbing
3	**(b)** from land to sea at night	Sea is warmer at night
4	**(b)** negative charges	Heavier water droplets
5	**(d)** rises, creating low pressure	Cooler air rushes in
6	**(a)** nothing of the chair remains	Matter is made of particles
7	**(b)** become weaker	Particles leave fixed positions
8	**(b)** Its particles are farther apart than those in water	Ice is an exception
9	**(d)** copper and tin	Example of alloy
10	**(b)** sea water	Salt dissolved evenly
11	**(d)** a homogeneous mixture	Uniform throughout
12	**(b)** salt in water	Solid dissolved in liquid

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Force due to action of muscles (1). Walking and lifting (1).	1+1
14	Pressure inside our body is equal to outside (1). So the pressures balance (1).	1+1
15	Gaps between particles (1). Solid state (1).	1+1
16	Substance of two or more elements combined chemically in a fixed ratio (1). Water or sodium chloride (1).	1+1
17	Solubility of most solids increases (1). That of gases decreases (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	kg is a unit of mass (1). Weight is a force measured in newtons (1). Correct statement: mass is 10 kg (1).	1+1+1
19	Land heated, warm moist air rises (1). Low pressure forms; cooler air rushes in (1). Water vapour condenses forming clouds, rain and strong winds (1).	1+1+1
20	Made of sodium and chlorine in 1 : 1 ratio (1). Evaporation is a physical process (1). Cannot split it into elements (1).	1+1+1
21	Stirring (1). Temperature (1). Size of solute particles (1).	1+1+1
22	Salt solution has higher density (1½). Egg is less dense than that solution (1½).	1½+1½
	OR: Add sugar to cold water until saturated (1). Warm the water and more sugar dissolves (1). So solubility increases with temperature (1).	1+1+1
23	Surfaces have minute irregularities (1). They lock into each other in contact (1). This opposes motion (1).	1+1+1
	OR: Mass = 50 ÷ 10 (1½). 5 kg (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	Definition (1). Move from rest (1). Change speed (1). Change direction (1). Change shape (1).	1+1+1+1+1
	OR: Spring stretches under load (1). Scale in newton (1). Least count = difference between marks ÷ number of divisions (1). Example 1 N ÷ 5 = 0.2 N (1). Reading with a numerical example (1).	1+1+1+1+1
25	Ice – close and regular (1). Water – close and mobile (1). Vapour – far apart (1). Melting and freezing (1). Boiling and condensation (1).	1+1+1+1+1

	OR: Syringe with air compresses (2). Syringe with water does not (1). Solid blocks cannot be compressed (1). Explain by interparticle space (1).	2+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET C

Chapters 5–9: **Exploring Forces, Pressure, Winds, Storms and Cyclones, Particulate Nature of Matter, Nature of Matter: Elements, Compounds and Mixtures, The Amazing World of Solutes, Solvents and Solutions** | Class 8 Science |

Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 12 × 1 = 12

Q.	Correct Option	Brief Reasoning
1	**(c)** mass	Matter does not change
2	**(b)** gravitational force	Acts downwards
3	**(a)** 270 km/h	As given in the chapter
4	**(d)** from sea to land during the day	Land heats faster
5	**(d)** its particles move freely in all directions	Negligible attraction
6	**(a)** particles of ink and water mix	Particles move and intermingle
7	**(a)** 1538 °C	Strong particle attraction
8	**(b)** elements or compounds	Two types
9	**(c)** Fruit juice	Contains many substances
10	**(c)** magnesium oxide	A compound
11	**(a)** salt	Dissolved substance
12	**(b)** kg/m³	Also g/cm³ in practice

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
13	Can attract or repel (1). Acts without contact (1).	1+1
14	Larger area reduces pressure (1). Same weight is spread out (1).	1+1
15	Matter is made of extremely small particles (1). There are spaces between them and forces between them (1).	1+1
16	Uniform: sea water, air (1). Non-uniform: muddy water, sand and iron filings (1).	1+1
17	Maximum amount of solute that dissolves in a fixed quantity of solvent (1). At a given temperature (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
18	It rises and slows down (1). Stops momentarily at top (1). Then falls with increasing speed due to gravity (1).	1+1+1
19	Liquid pressure increases with depth (1½). Bottom must withstand large horizontal pressure (1½).	1½+1½
20	Aluminium: closely packed regular (1). Glycerin: close but irregular (1). Methane: far apart (1).	1+1+1
21	Saturated: maximum dissolved (1). Unsaturated: can dissolve more (1). Dilute: small solute amount (1).	1+1+1
22	Start: kicking a football (1). Change shape: pressing a balloon (1). Change direction: bat hitting ball (1).	1+1+1
	OR: Least count = 1 ÷ 10 = 0.1 N (1½). Reading = 4 + 0.3 = 4.3 N (1½).	1½+1½
23	Area 225 cm² (1). Force about 2250 N (1). Equal to weight of 225 kg (1).	1+1+1
	OR: Stay indoors or in a car (1). Avoid tall trees and metal objects (1). Crouch low in open area, do not lie flat (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
24	Acts in all directions (1). Increases with height of column (1). Dam base broad (1½). Water tank at height (1½).	1+1+1½+1½
	OR: Warm air rises, low pressure (1). Cool air rushes in (1). Condensation forms clouds (1). Charges produced in clouds and lightning flashes (1). Thunder follows (1).	1+1+1+1+1
25	Composition (1). Properties (1). Separation (1). Formation (1). Example (1).	1+1+1+1+1
	OR: Mixing, magnet test (1). Heating (1). Black mass formed (1). Not magnetic (1). Iron + sulfur → iron sulfide (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.